

AI- POWERED WASTE MANAGEMENT SYSTEM FOR IKEA



INDIVIDUAL ASSIGNMENT



Proposed to:
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Big Data & AI.
Trinity Business School

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Declaration

I declare that this work has not been submitted as an exercise for a degree at this or any other university and it is entirely my own work.

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Signed: Rhythm Chaudhary

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Date: 7 April 2025

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IKEA, a global name in the furnishings industry, faces challenges in cutting down waste from unsold goods, packaging & surplus inventory. Thus, being a victim of unbalanced supply chains & ineffective demand forecasting.

IKEA's CEO has launched an AI-driven initiative to enhance sustainability by minimizing waste across the supply chain and in-store operations. Aligned with the company's broader data strategy, this initiative integrates AI, blockchain to drive efficiency, transparency and environmental responsibility.

2.1 CEO's DIRECTIVE

“The aim is to develop an intelligent waste minimizing system, enhancing the material efficiency, optimizing the management of product lifecycle and fortifying recycling initiatives.” - As directed by CEO

The proposed AI-powered waste Management System (AWMS) will be able to track waste materials, forecast waste trends and suggest 3Rs strategies by utilizing machine learning (ML) and deep learning models. The integration of real-time decision-making through a smooth exchange between AWS cloud services and IKEA's blockchain-supplied supply chain using AWMS establishes sustainable inventory management.

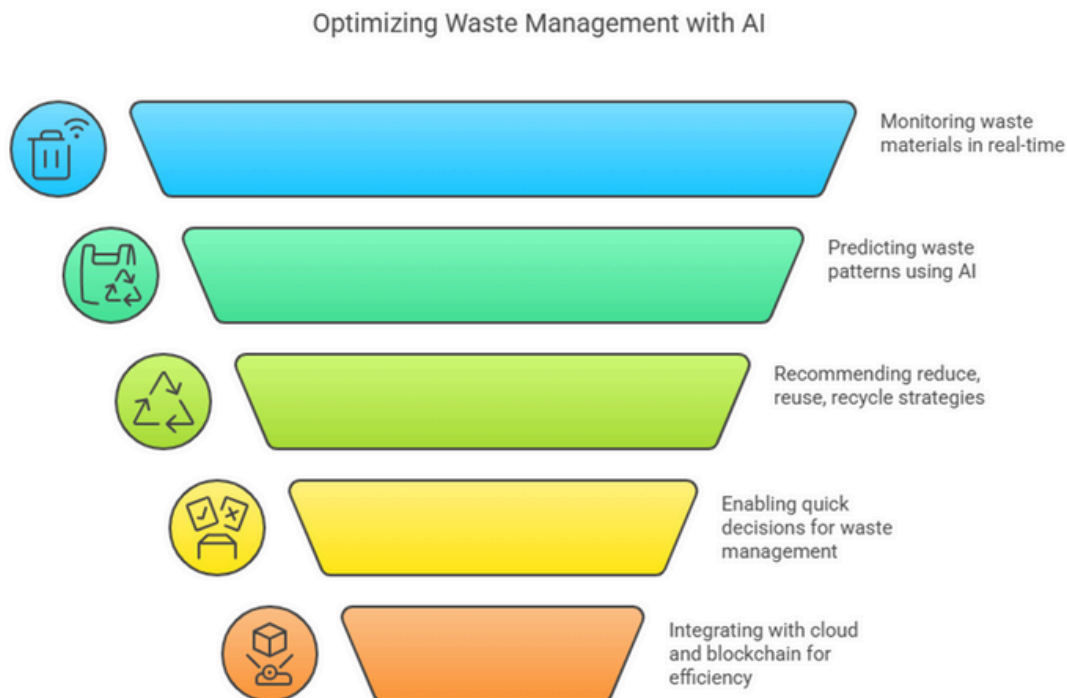


Fig2.1. Waste Management Optimization using AI

2.2 STRATEGIC SIGNIFICANCE

Companies like IKEA must show unambiguous, data-supported pledges to sustainable practices in order to comply with global environmental standards, consumer expectations, and corporate responsibility imperatives.

To accomplish this purpose, AWMS will be a key AI-enabled operational pillar that uses data insight to enable circular economy objectives.

It is referred as the foundational phase, since it develops the framework of the project. It is built into key deliverables such as proof of concept (PoC), proof of value (PoV), risk assessment, and stakeholder engagement.

3.1 IDEATION

Inputs were taken from supply chain managers, sustainability experts & data scientists before initiating the whole process. The following use cases were identified:

- a. Historical data-based waste forecasting by product line and store.
- b. Identification of types of waste material (such as textile, wood and plastic) through computer vision models

3.2 RISK ASSESSMENT

Category	AWMS System Area	Risk Description	Mitigation Strategy
High Risk	Waste Forecasting & Inventory Optimization	AI impacts critical supply and waste decisions	Human oversight, model audits, logging
High Risk	3Rs Recommendations & Reusability Scoring	Risk of bias or unclear automated suggestions	Use XAI, run fairness checks, diversify training
Limited Risk	Smart Waste Sorting (Computer Vision)	Possible misclassification of materials	Accuracy monitoring, manual override
Limited Risk	Blockchain Lifecycle Traceability	Data sensitivity and traceability concerns	GDPR compliance, anonymization, access control
Minimal Risk	Sustainability Reporting Dashboards	Used only for internal insights and visual reporting	Standard IT governance and system audits

Table 1. Risk assessment overview (EU AI Risk Framework)

3.3 PROOF-OF-CONCEPT

The Proof-of-Concept Plan helps us get a clear picture of the business's needs which is to:

- 1.Reduce wasteful overproduction and excess inventory
2. Minimize the environmental impact and disposal expenses.
- 3.Boost the transparency and efficiency of the supply chain.
- 4.Enhance the customer experience by using sustainable business practices

We'll begin by defining key waste metrics, particularly where the waste is being generated from, set sustainability goals that aligns with company's commitment to reduction in carbon footprint and then we'll choose 2-3 Ikea warehouses for the implementation.

3.3.1 DATA SOURCES

The following high-quality and diverse data streams will be used to power the AI models:

- Sales and Inventory Data: Real-time sales records, restocking rates, and demand-supply mismatch
- Logistics and Warehouse Data: Supplier delivery schedules, losses in shipment, and returns.
- Data on Removal of Consumer Discard: Unsold food, damaged goods, and disposal in the bin.
- IoT Sensor Data: Deployed in smart bins, recycling centres and with waste segregation units, to monitor volume and types of waste in real-time.
- Blockchain Product Lifecycle Data: Provides traceability of materials and supply chain checkpoints

3.3.2 MODEL DEVELOPMENT

For the models, we'll develop three core AI modules:

- Predictive Waste Forecasting Model: ML algorithms (ARIMA, LSTM, etc) that are trained on historical and real-time inventory data to help manage stock and prevent overproduction.
- Smart Waste -Sorting System Using AI: Apply Computer vision and CNN - Convolutional neural networks to classify discards by recyclability to better separate the waste streams.
- Demand Forecasting Model: Dynamic training on product ordering: on predicted consumer trends and seasonality to minimize excess inventory.

Once AI models are calibrated, real-time insights will be delivered via a waste tracking dashboard, with experimental deployments guiding refinements to improve accuracy, efficiency, and algorithm performance.

Task	Metric
Material Classification	IoU (Intersection over Union), Precision, Recall
Forecasting Accuracy	RMSE (Root Mean Square Error), MAE (Mean Absolute Error)
Alert Efficiency	Real-time alert latency < 3 seconds
Business Impact	% Waste Reduction, Cost Savings, Recyclability Rate, Staff Feedback

Table 2. Metrics Evaluation Summary

3.3.3 IMPACT MEASUREMENT

The final stage is impact measurement. Success will be measured in terms of percentages of waste saved, cost savings and environmental benefits. Feedback will be taken from employee and customer for upgrading the usability and integration of AI.

If the POC (Proof of Concept) is successful, the solution will be implemented worldwide by IKEA which will optimize models and infrastructure for a seamless, AI-based waste management process in its stores around the globe.

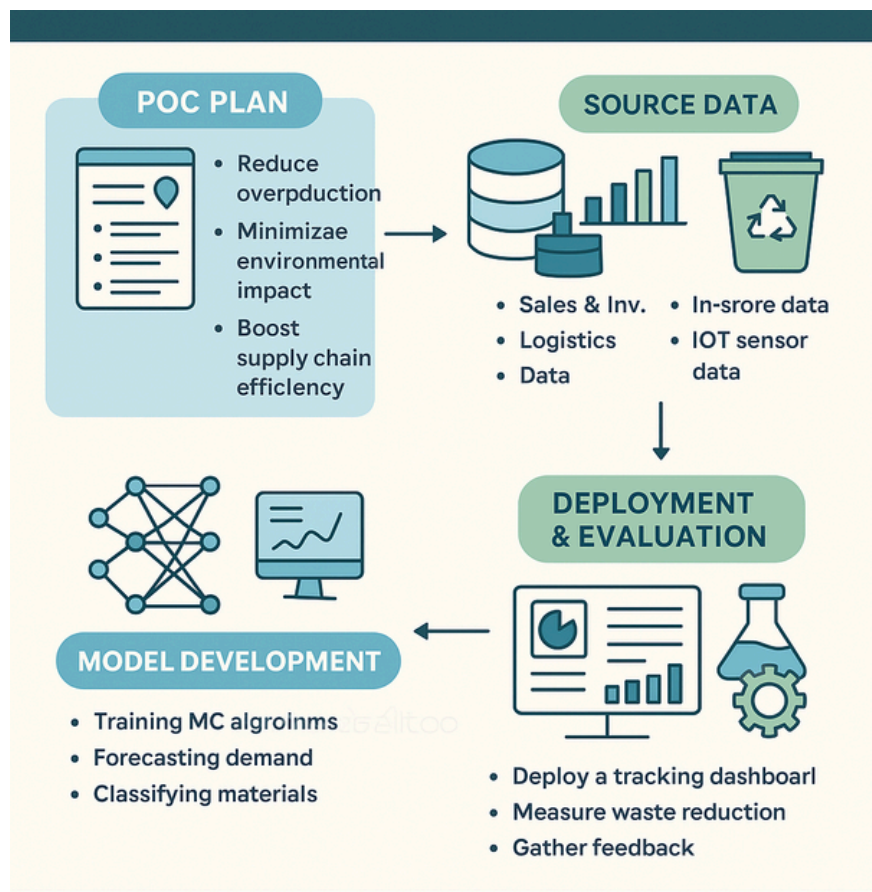


Fig 2. Diagrammatic Representation of PoC

3.4 PROOF-OF-VALUE

In IKEA's operations, the Proof of Value (POV) stage seeks to verify the observable advantages and return on investment (ROI) potential of the AI-powered Waste Management System (AWMS).

We're piloting an AWMS that is built to do three key things- cut unnecessary waste, save money, and make our operations greener and smarter.

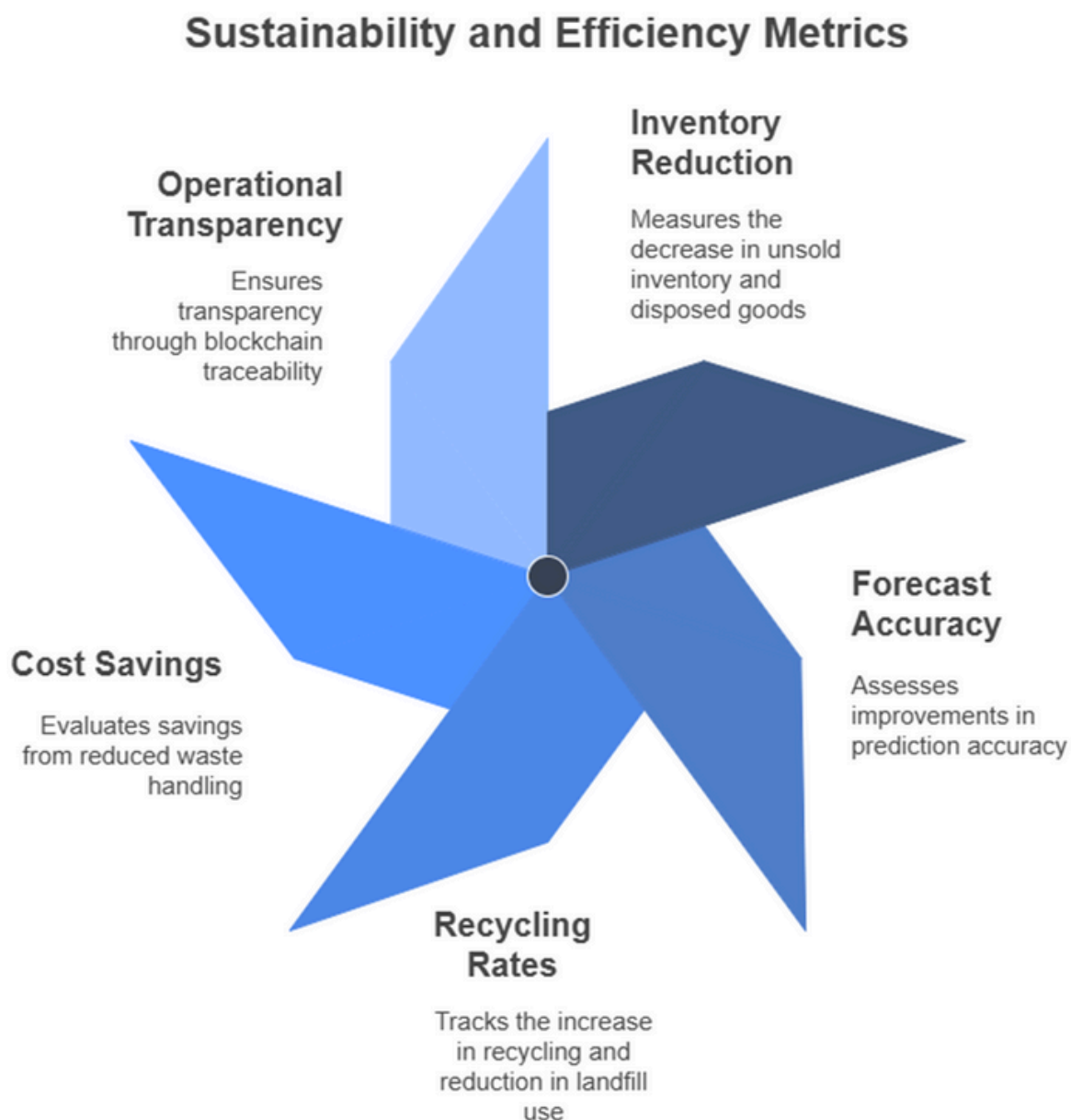
A PoV exercise will assess:

- Reduction in landfill waste volume- Target: 10% within first 3 months
- Improved product return-to-market ratio- Target: 5% increase via reuse
- Savings on operations due to improved material recovery

Operationally, the AWMS integrates waste-related data into IKEA's blockchain ecosystem to improve traceability & transparency of supply chain. This will help from start to finish material responsibility and ethical source verification.

Coordinating across geographies and business units is made easy by integration with AWS cloud services, which guarantees scalability in model deployment and centralized monitoring in real time. AWMS directly favours IKEA's commitment to reduce carbon footprint.

To keep it grounded and results-driven, we will start the execution with piloting AWMS across 2-3 locations. A baseline evaluation of present situation of waste in the industry will be executed by using historical, IoT sensor, inventory and logistics data. During the pilot, time series-based demand forecasting models and CNN classification systems will be deployed, and their performance will be compared to existing operations.



Among the Key Performance Indicators (KPIs) will be:

1. The percentage decrease in unsold inventory and discarded items
2. Enhancements in forecast accuracy (such as a decrease in RMSE)
3. Higher recycling rates and lower contributions to landfills
4. Reduced expenses due to less waste management and inventory write-offs

Furthermore, we will be evaluating model accuracy, decision latency and reliability of the system. A solution is only as good as it is usable, so we will gather feedbacks from the employees- store managers, warehouse teams & sustainability officers too, guiding real time improvements and ensuring alignment within stakeholders.

During the pilot, a specialized dashboard will be created to help in decision-making and visualize insights.

The POV's results will be combined into a comprehensive report that quantifies the AWMS's strategic, financial, and environmental value for IKEA management. IKEA will be able to lead the retail sector in AI-enabled sustainability and data-driven circularity if the system is effective in scaling globally.

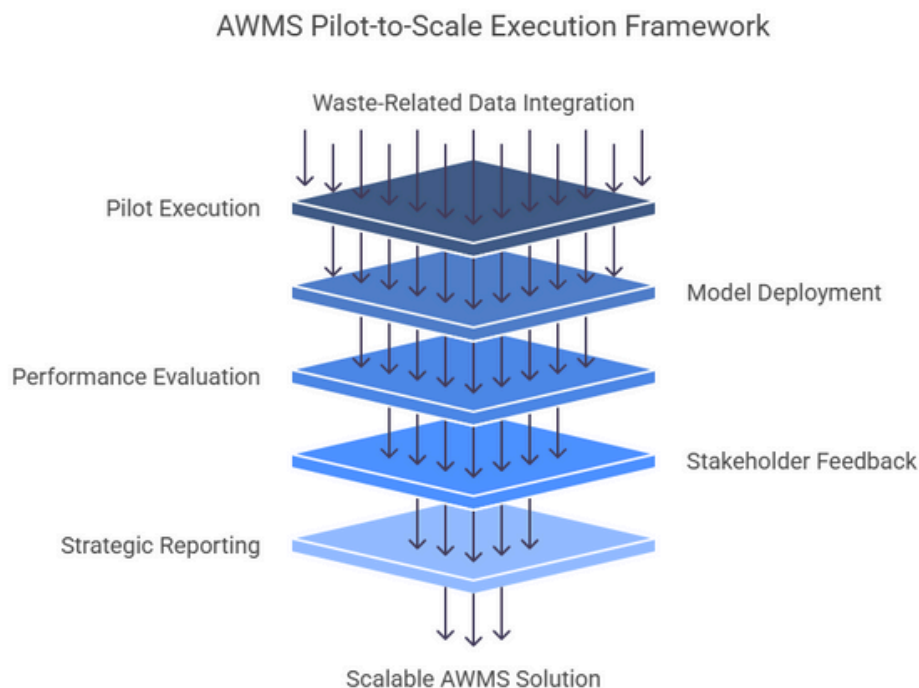


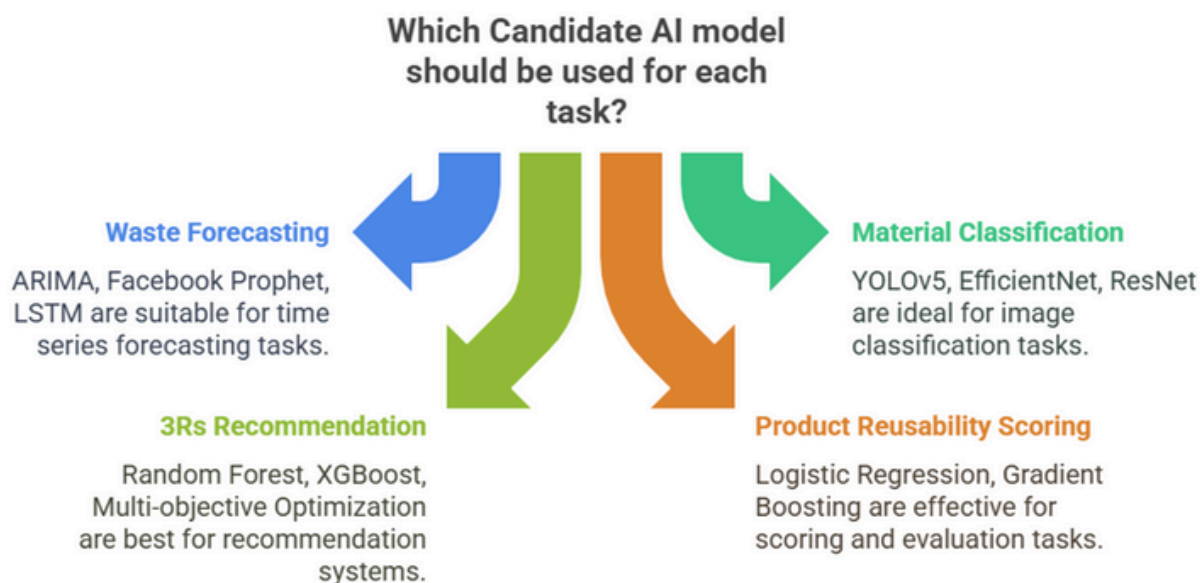
Fig 4: A strategic roadmap from pilot integration to actual adoption

3.4.1 DATA SOURCE

DATA TYPE	SOURCE	FREQUENCY	USE CASE
Waste Logs	Distribution <u>Centers</u>	Daily	Forecasting
CCTV & Visual Waste Bins	Video/Image	Real-Time	Classification
Product Lifecycle DB	Internal ERP	Weekly	Lifecycle Optimization
Blockchain Product ID	Supply Chain API	Real-Time	Material Traceability
Sales/Returns	POS Systems	Daily	Reverse Logistics

Table 3. Relevant Data Sources

3.4.2 MODEL APPROACH



Model performance will be validated using standard metrics (F1-score, RMSE, AUC) and cross-validation techniques.

3.4.3 TEAMS, TOOLS, BUDGET

CATEGORY	DETAILS
Core Tools	AWS SageMaker, Power BI, Tableau, Jupyter,
	Python, PyTorch
Team Members	2 Data Scientists
	1 ML Engineer
	1 AWS Cloud Engineer
	1 Blockchain Developer
	1 Sustainability SME
	1 Project Manager
Initial Budget	PoC Pilot Facility Setup: €100,000
	ML & Cloud Infrastructure: €150,000
	Personnel and Training: €80,000
	Total Estimated for Development: €330,000

Once the AI models have been developed and validated successfully in PoC, the next step is the operational launch of the AWMS in selected IKEA's pilot warehouses. There needs to be a phase where the AI tools are put into real infrastructure, tested in a controlled manner, personnel are trained, operations are scaled while ensuring reliability and business continuity.

4.1 DEPLOYMENT OF MODELS & INTEGRATING INFRASTRUCTURE

During the deployment phase, the trained AI models (waste forecasting, smart sorting, 3Rs recommendation) will be packaged up and converted to production-ready APIs. We will be hosting on AWS SageMaker and orchestration of containers will happen via AWS EKS for scalable and containerized operations.

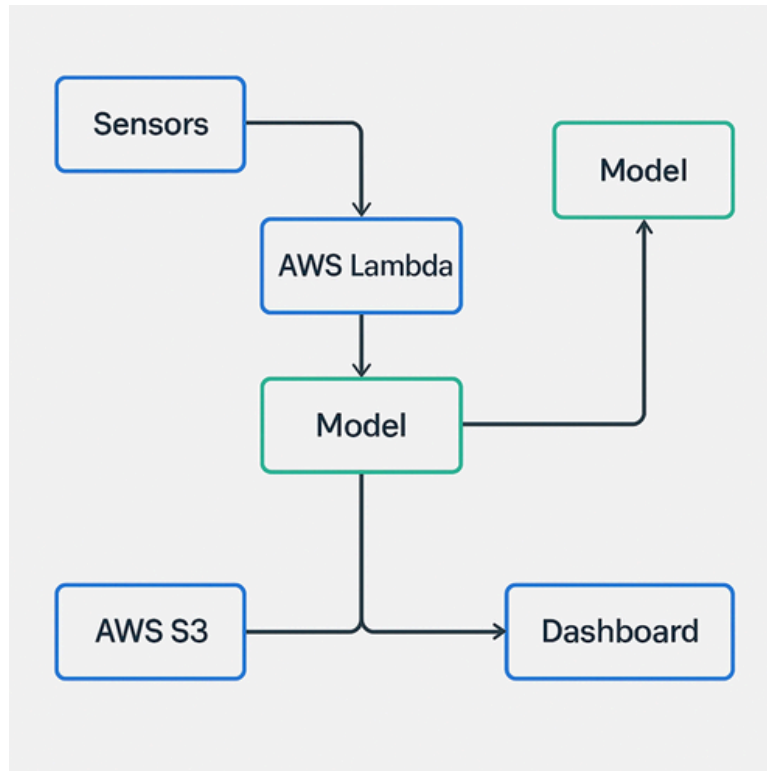


Fig. 5

- Inference Mode: Real time and batch predictions as per task basis ,for instance, sorting real-time, forecasting hourly.
- IoT Integration: AWS Lambda will be used through which we will have predictions, when real time sensor data from waste bins and warehouses will stream into AWS Kinesis.
- Data Storage: The main storage layer for structured and image-based inputs will be AWS S3 and Redshift.

The system will follow a microservices architecture, allowing independent scaling of modules like the computer vision service and forecasting engine.

4.2 INTEGRATION WITH IKEA'S DIGITAL ECOSYSTEM

Our AWMS will adjust without any hassle into IKEA's existing interface with minimal disruption to its operations and will ensure smooth adoption.

It will integrate directly with IKEA's enterprise resource planning (ERP) systems, allowing it to sync with inventory levels and tools for managing the lifecycle of a product.

Furthermore, as the system will connect to IKEA's blockchain network where APIs will be used to retrieve and validate smart contracts and origin of product data at important lifecycle checkpoints. The deployment utilizes IKEA's existing AWS infrastructure, GDPR-compliant anonymization policies, and cloud-native analytics tools (Power BI/Tableau).

4.3 PILOT TESTING & SCALING

The process will begin with the pilot warehouse used for trials in PoC. AWMS Facilities will be compared to the sites using traditional methods of tracking wastes and to achieve this A/B Testing will be conducted.

Since, this is a phased rollout method, there will be 4 phases- In Phase 1, AI models will be deployed to those 2-3 warehouses which are already used in PoC. In Phase 2, A/B Testing will be conducted for almost 3 months and KPI's will be evaluated. In Phase 3, feedbacks will be incorporated and fine tune models and lastly in Phase 4, the models will be gradually rolled out to more warehouses based on performance.

Success Criteria for Scaling



Fig 6. Indicators for Successful Scaling

4.4 TRAINING & MANAGING THE CHANGE

IKEA will employ a human-centric change management approach backed by thorough training to guarantee effective adoption.

Plan for Staff Training:

1. Operations and warehouse team's onboarding sessions.
2. Walkthroughs of the dashboard and live demos
3. SOPs and training materials incorporated into IKEA's internal learning management system

Measures for Change Management:

1. Engage warehouse employees and sustainability leaders in early feedback loops.
2. At each location, choose local AWMS "champions" to support adoption.
3. Surveys should be conducted often both during and after the rollout.

4.5 MONITORING & SLA's

Clear SLAs for model uptime ($\geq 99.5\%$), real-time latency (≤ 3 seconds), and dashboard availability (24/7) need to be defined. Moving forward, a dedicated DevOps team will be assigned to monitor the AWS services deployed and guarantee system performance through continuous tracking, and the retraining of models on a schedule.

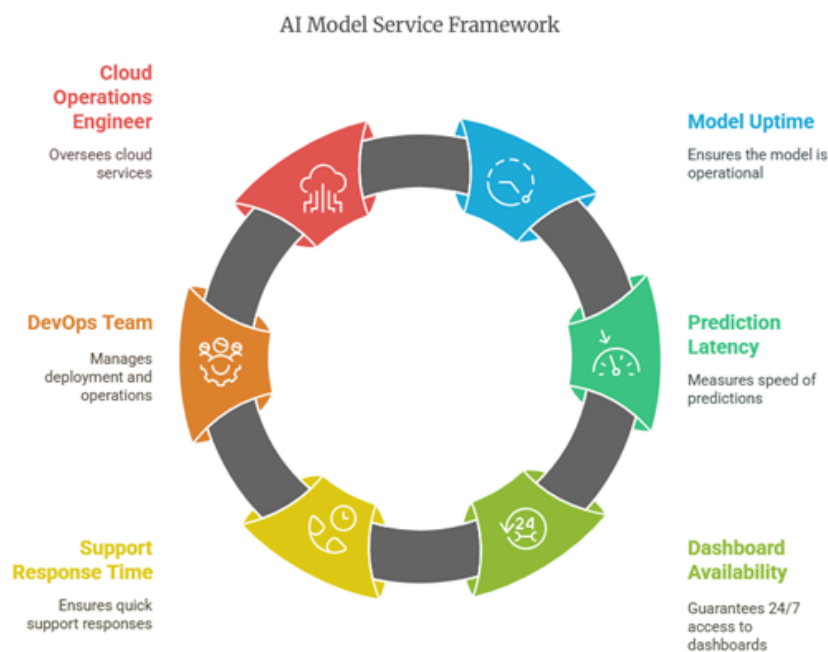


Fig 7. Monitoring Technique & SLA Framework

The maintenance stage makes sure our AWMS does not reach the tendency to be unreliable, unethical and ineffective.

It involves processes like continuous monitoring, updates, feedback integration and compliance to stay in line with ever-changing requirements, standards, and AI improvements.

5.1 PERFORMANCE MONITORING AND CONTINUOUS IMPROVEMENT

AWMS will be subjected to monitoring through automated processes tracking the model drift, forecast error, system uptime, etc. Moreover, a dedicated field-specific DevOps/ML team will be responsible for retraining, version control and CI/CD based update of models to ensure consistent performance with minimal downtime.

5.2 INCORPORATE FEEDBACKS & SYSTEM ADAPTATION

User feedback would be collected through dashboard surveys, staff review sessions, and usage analytics, which will help optimize the use of dashboards, enhance AI recommendations and system alerts to better align with real-world operations.

5.3 GOVERNANCE, ETHICS, COMPLIANCE



AWMS is a significant milestone in IKEA's digital and sustainability roadmap. The system builds on its data strategy using AI, AWS and blockchain to impact waste reduction, lifecycle efficiency and a circular economy. The Proof of Concept has confirmed technical feasibility and business value, laying the foundation for full deployment and long-term continuance of the solution.

Key Recommendations:

Stage AWMS Deployment: Deploy to priority sites based on lessons learned from the PoC.

**Scale AWMS Deployment**

Deploy to priority sites based on lessons learned from the PoC.

Examine the convergence of sustainable design and AI-driven reverse logistics.

**Invest in strengthening MLOps**

Investments to be made in robust monitoring, retraining and deployment pipelines

**Enhance Feedback Loops**

Genuine feedbacks with asurity should be taken from staffs to improve usability & real time alignment

**Maintain Ethical AI practices**

Uphold transparency, fairness, and GDPR compliance.

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Generative AI Declaration:

Although artificial intelligence (AI) tools were utilized in the preparation of this assignment, their function was restricted to aiding with presentation clarity, structure, and formatting. All of the main concepts, analysis, insights, and arguments made in this work are derived solely from my own knowledge, growth, and critical thinking throughout the course.

I've always made it a point to maintain academic integrity by making sure that the work shows my sincere effort and understanding. AI was not used to help with content creation or conceptual development; its only goal was to improve communication. In the spirit of accountability and openness, I share this.